Quantum Physics - Notes - v0.2.0

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Preface

Every theory section in these notes has been taken from the sources:

- Quantum Mechanics. [1]
- Fondamenti di fisica atomica e quantistica. [2]
- Modern Quantum Mechanics. [3]

About:

GitHub repository

These notes are an unofficial resource and shouldn't replace the course material or any other book on quantum physics. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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1 From Wave-Particle Duality to Quantum States

1.1 Experimental Motivation: Wave-Particle Duality

The purpose of this section is not to give a history of the development of quantum mechanics, but rather to discover what kind of mathematical theory we will eventually need to describe the phenomena that we will discuss in these notes.

We begin from a very basic question: "***what is light?***". Classically, there are two very different possibilities:

1. A **Particle** is something localized. At a given instant it **occupies a particular position**, **follows a trajectory**, and **arrives somewhere** as an individual object.
2. A **Wave**, on the other hand, is **spatially extended**. Different portion of the wave can overlap and interfere.

Historically, both descriptions were proposed for light. But for our purposes, the important question is not who proposed what. The important question is: *how can an experiment distinguish wave behaviour from particle behaviour?* In other words, definitions alone cannot choose between the two descriptions. So we need an **experimental criterion** that allows us to distinguish between the two behaviors.

⚖ What phenomenon is characteristic of waves? The wave vs. particle descriptions of light

Suppose we have light traveling from a source toward some obstacle and then onto a screen. If light behaved only like classical particles, the simplest picture would be:

$$\text{source} \to \text{particle trajectories} \to \text{screen}$$

Each particle follows some path, reaches a particular location, and contributes there.

But **waves behave differently** because different contributions can be added together. For two waves:

$$\psi_1, \quad \psi_2$$

The **combined disturbance**[1] is:

$$\psi_{\text{tot}} = \psi_1 + \psi_2$$

This simple operation has a crucial consequence: the two waves may reinforce each other or cancel each other. That phenomenon is called **Interference**.

[1] The **total/resultant wave** produced when the two waves are present at the same place and time.

So already we have the first important distinction between waves and particles:

- **Particles** are localized objects.
- **Waves** are superposable[2] and extended disturbances.

At this point, "superposition" is still the familiar classical-wave idea.[3] We are not yet stating the quantum superposition principle. Also, we have seen that the **interference phenomenon** is a characteristic of waves, but we have not yet seen any experimental evidence for it.

Diffraction and interference as a signatures of waves

Interference is often observed experimentally through a **Diffraction** pattern. It is one of the **most convenient experimental manifestations of interference**. In general, interference is the general wave phenomenon, and diffraction is a particular wave phenomenon in which a wave spreads after an aperture/obstacle, and the resulting diffraction pattern is produced by interference. So we can have interference without diffraction, for example two independent water waves meeting each other.

Deepening: What does a "wave" actually have?

Before talking about light, think about something more intuitive: a **wave on a rope**. Imagine a rope that is fixed or held at the ends. If we move one point up and down, the disturbance propagates along the rope. The rope itself does **not** travel along with the wave. Each small piece of rope mainly moves up and down around its equilibrium position, while the **wave pattern** propagates horizontally.

Mathematically, at each point x along the rope, it has some displacement from equilibrium:

$$y(x,t)$$

Which is the description of the wave, where y is the vertical displacement of the rope at position x and time t ($y = 0$ is the equilibrium position). At one instant maybe $y = 2$ cm, and somewhere else $y = -1$ cm.

Thus, the quantity y is the **wave amplitude** at that point and instant. It indicates how far the rope is displaced from its equilibrium position.

For the case of **light**, instead of writing $y(x,t)$, we can think about:

$$E(x,t)$$

So the "amplitude" is the **value of the oscillating wave quantity**.

[2] They can be added together. More precisely, in a linear wave system, if ψ_1 and ψ_2 are allowed waves, then $\psi_1 + \psi_2$ is also an allowed wave.

[3] The sum of two waves is still a wave.

Deepening: What happens when two waves meet?

This is the defining ingredient behind interference. Suppose two waves arrive at the same place. Their amplitudes add:

$$E = E_1 + E_2$$

This is the superposition principle for classical waves. Now imagine at instant:

$$E_1 = +1, \quad E_2 = +1$$

Then:

$$E = +2$$

They reinforce each other. But perhaps instead:

$$E_1 = +1, \quad E_2 = -1$$

Then:

$$E = 0$$

They cancel each other. That is interference. So the fundamental thing that interference is **the amplitude**: $E_1 + E_2$.

Suppose light hits a screen. We don't see the electric field E directly:

$$E = +3, \quad E = -2, \quad E = +1, \quad \dots$$

Because the electromagnetic field oscillates extremely rapidly, and our eyes cannot follow it. Instead, the screen receives **energy** from the light.

- A region receiving more energy appears brighter.
- A region receiving less energy appears darker.

The quantity describing this is called the **Intensity** I, the **energy delivered per unit area per unit time**, roughly speaking.

For a light wave, intensity is proportional to $\propto$ the square of the wave amplitude:

$$I \propto |E|^2$$

Why square the amplitude? One immediate reason the square makes sense is that amplitude can have a sign (e.g., $E = +1$ or $E = -1$), but **energy cannot be negative**. Squaring guarantees that the intensity is always positive. The sign is important for **interference**, but the final measured energy/intensity is always positive. Thus:

1. Add the amplitudes of the waves to get the total amplitude.
2. Square the total amplitude to get the intensity.

Example 1: Interference of two waves

Imagine each wave alone would have amplitude magnitude 1. If only wave 1 existed:

$$E_1 = 1$$

So its intensity would be proportional to:

$$I_1 \propto 1^2 = 1$$

Similarly, if only wave 2 existed:

$$E_2 = 1$$

So its intensity would be proportional to:

$$I_2 \propto 1^2 = 1$$

We might naïvely think that two waves together should therefore give:

$$I = I_1 + I_2 = 1 + 1 = 2$$

But that is **not necessarily true**. We must first add their amplitudes.

- **Case A: the waves reinforce each other**. If:

  $$E_1 = 1, \quad E_2 = 1$$

  Then:

  $$E = E_1 + E_2 = 1 + 1 = 2$$

  So the intensity is:

  $$I \propto |E|^2 = 2^2 = 4$$

  We have **constructive interference**.

  And if we had:

  $$E_1 = -1, \quad E_2 = -1$$

  Then:

  $$E = E_1 + E_2 = -1 - 1 = -2$$

  So the intensity is:

  $$I \propto |E|^2 = |-2|^2 = 2^2 = 4$$

  Again, we have **constructive interference**, even though the amplitudes are negative. The sign of the amplitude does not matter for the intensity.

- **Case B: the waves cancel each other**. If:

  $$E_1 = 1, \quad E_2 = -1$$

  Then:

  $$E = E_1 + E_2 = 1 - 1 = 0$$

  So the intensity is:

  $$I \propto |E|^2 = 0^2 = 0$$

  So the screen is dark there. This is **destructive interference**.

Let's generalize the previous example. Imagine one sinusoidal wave reaching some point P:

$$E_1(t) = E_0 \cos(\omega t)$$

Its average intensity is proportional to the square of the amplitude:

$$I_1 \propto E_0^2$$

So far, nothing strange. Now suppose **two waves** reach the same point P. Because ordinary wave equations are linear, the physical field is the sum:

$$E(t) = E_1(t) + E_2(t)$$

This is the classical **superposition principle for waves**. But intensity is not proportional to E, but rather to the **square** of the resulting amplitude:

$$I \propto |E|^2 \implies I \propto |E_1 + E_2|^2$$

Expanding:

$$|E_1 + E_2|^2 = |E_1|^2 + |E_2|^2 + 2\Re\left(E_1^* E_2\right)$$

The last term $2\Re\left(E_1^* E_2\right)$ is called the **Inference Term**. This term is the reason why:

$$|\psi_1 + \psi_2|^2 \neq |\psi_1|^2 + |\psi_2|^2$$

Or in words, the intensity of two waves together is not equal to the sum of their individual intensities. The inference term can be:

- **Constructive Interference** if the phases agree:

  $$2\Re\left(E_1^* E_2\right) > 0$$

  The two contributions reinforce each other. We have a **maximum** in the intensity.

- **Destructive Interference** if the phases disagree:

  $$2\Re\left(E_1^* E_2\right) < 0$$

  The contributions oppose each other and may cancel. We have a **minimum** in the intensity.

So an alternating pattern of maxima and minima is a signature of interference, and it is a signature of wave-like behavior. Therefore, observing such a diffraction/interference pattern provides an experimental criterion for identifying the wave-like nature of light.

> **Deepening: The interference term**
>
> Let's remove for a moment the absolute value from each term. Then the intensity is still proportional to the square of the amplitude:
>
> $$I \propto (E_1 + E_2)^2$$

Expanding it exactly like ordinary algebra:

$$(E_1 + E_2)^2 = E_1^2 + E_2^2 + 2E_1E_2$$

The last term $2E_1E_2$ is the **Interference Term** in the simplest real-valued case. It is really **important** because without it, we would just have:

$$E_1^2 + E_2^2$$

Which is just "intensity from wave 1 plus intensity from wave 2". The extra term tells us what happens when the two waves meet.

For example, if $E_1 = 1$ and $E_2 = 1$, then the intensity is:

$$I \propto (1+1)^2 = 4$$

But separately, the intensity from each wave is:

$$I_1 + I_2 = 1^2 + 1^2 = 2$$

The extra term $+2$ comes from the interference term:

$$2E_1E_2 = 2 \cdot 1 \cdot 1 = 2$$

So the waves reinforce each other. Now take $E_1 = 1$ and $E_2 = -1$. Then the intensity is:

$$I \propto (1-1)^2 = 0$$

But separately, the intensity from each wave is:

$$I_1 + I_2 = 1^2 + (-1)^2 = 1 + 1 = 2$$

The extra term -2 comes from the interference term:

$$2E_1E_2 = 2 \cdot 1 \cdot (-1) = -2$$

So the interference term is simply what can either increase the intensity or decrease it.

Now, why did we previously write the interference term as $2\Re\left(E_1^* E_2\right)$ instead of $2E_1E_2$? Because for light waves we often represent the amplitude using complex numbers. Then the correct version of the same algebra becomes:

$$\begin{aligned}
|E|^2 &= |E_1 + E_2|^2 \\
&\downarrow |z|^2 = z \cdot z^* = (x+iy)(x-iy) = x^2 + y^2 \\
&\downarrow \text{ if we assume } E_1 + E_2 = z \\
&= (E_1 + E_2)(E_1 + E_2)^* \\
&= (E_1 + E_2)(E_1^* + E_2^*) \\
&= E_1E_1^* + E_1E_2^* + E_2E_1^* + E_2E_2^* \\
&\downarrow E_1E_1^* = |E_1|^2, \quad E_2E_2^* = |E_2|^2 \\
&= |E_1|^2 + |E_2|^2 + E_1E_2^* + E_2E_1^*
\end{aligned}$$

Now the last two terms are complex conjugates of each other. Let:

$$z = E_1 E_2^*$$

Thus its complex conjugate is:

$$z^* = (E_1 E_2^*)^* = E_1^* \, (E_2^*)^* = E_1^* E_2$$

Therefore:

$$E_1 E_2^* + E_2 E_1^* = z + z^*$$

And for any complex number:

$$z = a + ib$$

We have:

$$z^* = a - ib$$

So:

$$z + z^* = a + ib + a - ib = 2a$$

But a is the real part of z, represented as $\Re(z)$. So we have:

$$z + z^* = 2\Re(z)$$

Therefore:

$$E_1 E_2^* + E_2 E_1^* = 2\Re\left(E_1 E_2^*\right) \equiv 2\Re\left(E_1^* E_2\right)$$

And conceptually it is just the same cross term as $2E_1 E_2$.

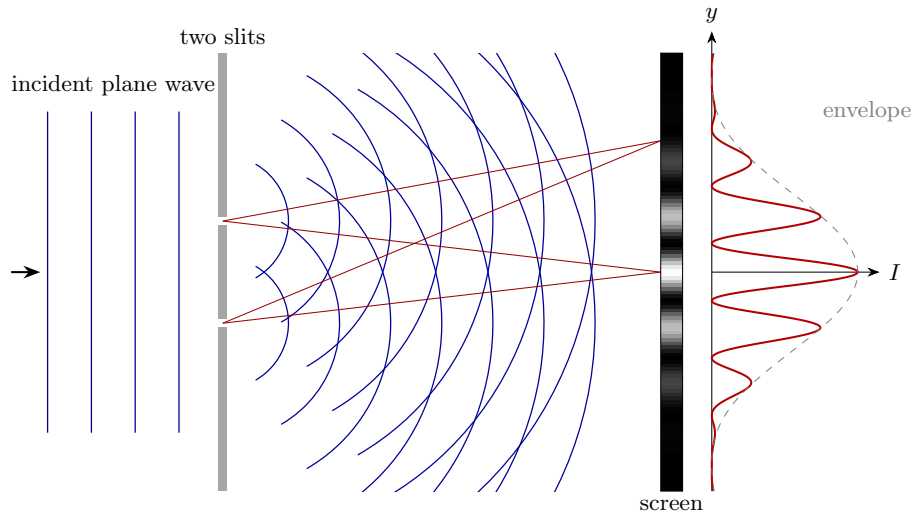


Figure 1: **Diffraction and interference in a two-slit experiment.** An incident plane wave passes through two slits; the emerging waves spread, overlap, and interfere, producing an intensity pattern on the screen modulated by a diffraction envelope.

Deepening

Some details about the Figure 1:

What is the "incident plane wave"? On the left we have several **parallel blue vertical lines**, those lines are **wavefronts of the same incoming wave**.

A wavefront is a set of points where the wave is in the same state of oscillation. We do not need to know particular details; visually, think of each blue line as one crest/front of the wave. Because these wavefronts are straight and parallel, we call it a **plane wave**. So "*incident plane wave*" simply means a wave arriving at the barrier with flat, parallel wavefronts. "*Incident*" just means **incoming**.

Finally, the arrow indicates that the wave propagates from left to right.

What are the "two slits"? The grey vertical object is a barrier with **two small openings**. Most of the wave is blocked by the barrier, only the portions reaching those openings pass through.

Here, the crucial observation is what happens **after** each slit. Instead of continuing only straight ahead, the wave **spreads out**. That's why the figure draws curved blue wavefronts coming from each slit.

This spreading is the **diffraction** part! Each slit effectively gives us a new spreading wave. And now we have precisely what we need fo **interference**: two waves reaching the same points on the screen (black rectangle on the right).

What happens between the slits and the screen? Pick some point P on the screen. There are two wave contributions reaching P: ψ_1 from the upper slit and ψ_2 from the lower slit. The red straight lines in the figure illustrate examples of those two paths.

The two contributions add:

$$\psi(P) = \psi_1(P) + \psi_2(P)$$

But depending on the point P, the two paths have different lengths. So the waves may arrive "*in phase*" (reinforce, constructive interference), "*out of phase*" (cancel out, destructive interference), or somewhere in between (weaker reinforcement or weaker cancellation). And that's why different points of the screen have different brightness.

Why is the center of the screen bright? Call the two slit-to-screen distances r_1 and r_2. At the exact center of the screen, by symmetry, the two distances are equal:

$$r_1 = r_2$$

Therefore the two waves have traveled the same distance, and they reinforce each other:

$$\psi_1 + \psi_2 \implies I \propto |\psi_1 + \psi_2|^2$$

So the intensity is large and the central bright band corresponds to constructive interference.

What are the dark bands? Move upward or downward along the screen. Now one slit is slightly closer than the other:

$$r_1 \neq r_2$$

At certain position, the difference in traveled distance causes one wave to arrive exactly opposite in phase to the other.

For example:

$$\psi_1 = A, \qquad \psi_2 = -A$$

Therefore:

$$\psi_1 + \psi_2 = A - A = 0$$

And hence the intensity is zero:

$$I \propto |\psi_1 + \psi_2|^2 = 0$$

So that location is dark, corresponding to destructive interference.

What is that red curve on the right? The horizontal direction of the red graph represents the **intensity** I measured on the screen, while vertically we move along the screen coordinate y.

So where the solid red curve extends far to the right, the intensity is large and the screen is bright. Where it comes close to zero $I \approx 0$, the screen is dark.

So the oscillating solid red curve is essentially a graph of $I(y)$, the intensity as a function of vertical position on the screen.

What is the "envelope"? Look at the **dashed red curve** labelled "envelope". It is **not** "*the level of interference*". Instead, it describes how large the interference peaks are allowed to become at different positions.

A more natural question is: ***why are the bright fringes near the center stronger than the bright fringes farther away?*** What is exactly what the dashed curve in figure is trying to show. The red solid curve $I(y)$ oscillates between bright and dark because of the **interference between the two slits**.

But notice also that the heights of the bright peaks are not all the same. Near the center, the peaks are large; farther away, they become smaller. The dashed curve simply joins the tops of those possible peaks. So the **envelope** is the smooth curve giving the

maximum possible intensity at each position on the screen (at each y). In other words, it is just a boundary for the interference pattern.

But why does the envelope exist? Because the slits are not mathematical points; they have a finite width. Even **one slit by itself** produces a diffraction (i.e., spreading) pattern on the screen. So each slit does not send the same wave strength equally in every direction. it sends most strongly approximately forward, around the center of the screen, and more weakly at larger angles.

Therefore, before we even combine the two slits, each slit already has something like: strong near the center, weaker farther away. That broad distribution is the **single-slit diffraction pattern**. Now, if we combine the two slits, the two effects happen simultaneously and each slit diffracts; then the waves from the two slits interfere and the final pattern is the product of the two effects:

- **Single-slit diffraction** gives the broad envelope.
- **Two-slit interference** gives the oscillating pattern within that envelope.

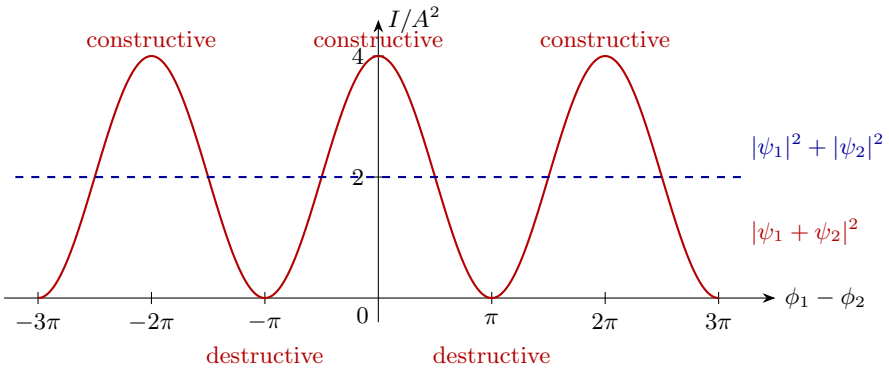


Figure 2: **The interference term.** The resulting intensity depends on the relative phase $\phi_1 - \phi_2$: constructive interference produces maxima, while destructive interference produces minima.

Electron diffraction

Now we take the criterion we just established for waves (i.e., the presence of interference)

$$\text{diffraction/interference} \Longrightarrow \text{wave-like behaviour}$$

And apply it to something that, classically, we would expect to behave as a particle: the electron.

The surprising experimental observation is that a beam of electrons, when sent onto a crystalline sample, produces a pattern that looks essentially like the diffraction pattern produced by light.

The idea. Before this experiment, the classical expectation was: electron behaves $\approx$ particle. *Why?* Because electrons have properties we normally associate with particles, they have mass, charge, momentum, and can form an electric current.

But now we perform an experiment designed to test for wave behaviour. We send electrons toward an ordered crystalline structure. And experimentally we observe, again, a diffraction pattern. And from the previous discussion, we know that it means:

$$\text{diffraction pattern} \Longrightarrow \text{interference} \Longrightarrow \text{wave-like behaviour}$$

So the immediate conclusion is that electrons behave like waves.

Why is this surprising? Because the same physical object (the electron) was already well described as a particle in other experiments. So we have:

$$\text{electron} \begin{cases} \text{particle-like in some experiments} \\ \text{wave-like in diffraction experiments} \end{cases}$$

This is already the beginning of **wave-particle duality**. However, we shouldn't jump ahead just yet. We'll see it in more detail later. At this point, all we have established is that an electron beam produces a wave-like diffraction pattern.

What exactly do we observe?

1. The electron source produces a collimated beam of electrons.
2. These electrons hit a crystal.
3. A crystal contains atoms arranged in a regular periodic structure.
4. The electrons interact with that periodic structure, and the detected electrons are not distributed uniformly everywhere. Instead, they **appear preferentially in specific directions**, giving a pattern of bright spots.

That is exactly the kind of spatial organization we expect from diffraction. The regular atomic arrangement plays, for electrons, a role analogous to a diffraction grating for light.

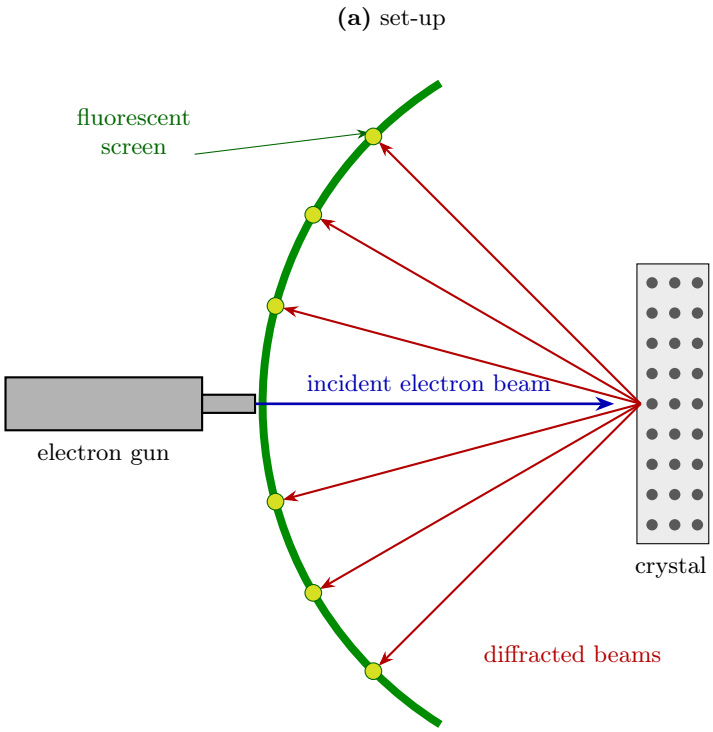


Figure 3: **Electron diffraction from a crystal.** A collimated electron beam produced by an electron gun is incident on a crystalline sample. The scattered electrons are detected on a fluorescent screen, where they form an ordered diffraction pattern with bright maxima separated by low-intensity regions.

In Figure 3, we show a schematic of the experimental set-up and the resulting diffraction pattern.

- In panel (a), the **electron gun** is the **source**. It emits a collimated beam of electrons **toward the crystal**. The crystal is not just a random obstacle: its atoms are arranged periodically, so it behaves like a diffraction structure for the electron beam.

  The **arrows** leaving the crystal represent the **diffracted directions**. They are not meant to say that we know a classical trajectory for each electron; they indicate the directions where the scattered electron signal is enhanced because of diffraction.

  Those **electrons then hit the fluorescent screen**. The screen converts the electron impacts into visible light: the electrons excite the fluorescent material, which then emits light.

- Panel (b) shows what we actually observe on that screen: a set of bright spots at specific positions. Those bright spots are the diffraction maxima. Between them, the signal is much weaker or absent. That ordered pattern is the important experimental result.

Single-particle regime

From the previous discussion, we observed that a beam of electrons produces a diffraction pattern after interacting with the crystal, that suggests wave-like behaviour. But there is still **an alternative explanation** we have **to exclude**: maybe the diffraction pattern is caused by interactions among many electrons.

Electrons are charged particles, so when many of them are present simultaneously they **can interact with one another**. So we need to find a way to eliminate that possibility, and the solution is to send only one electron at a time.

How do we do that? We **reduce the intensity** of the electron current emitted by the **electron gun** until successive electrons are separated by a sufficiently large time interval. So instead of having something like:

$$e^- \; e^- \; e^- \; e^- \; e^-$$

Inside the apparatus (i.e., in the region where the electrons are interacting with the crystal), we arrange approximately:

$$e^- \qquad\qquad e^- \qquad\qquad e^-$$

With a large delay between one electron and the next. So we reduce the current until one electron is sent at a time, with enough delay that individual arrivals can be resolved experimentally. This is what we call the **single-particle regime**.

In the next page (Figure 4), we show a schematic of the two regimes: single-particle and many-electron.

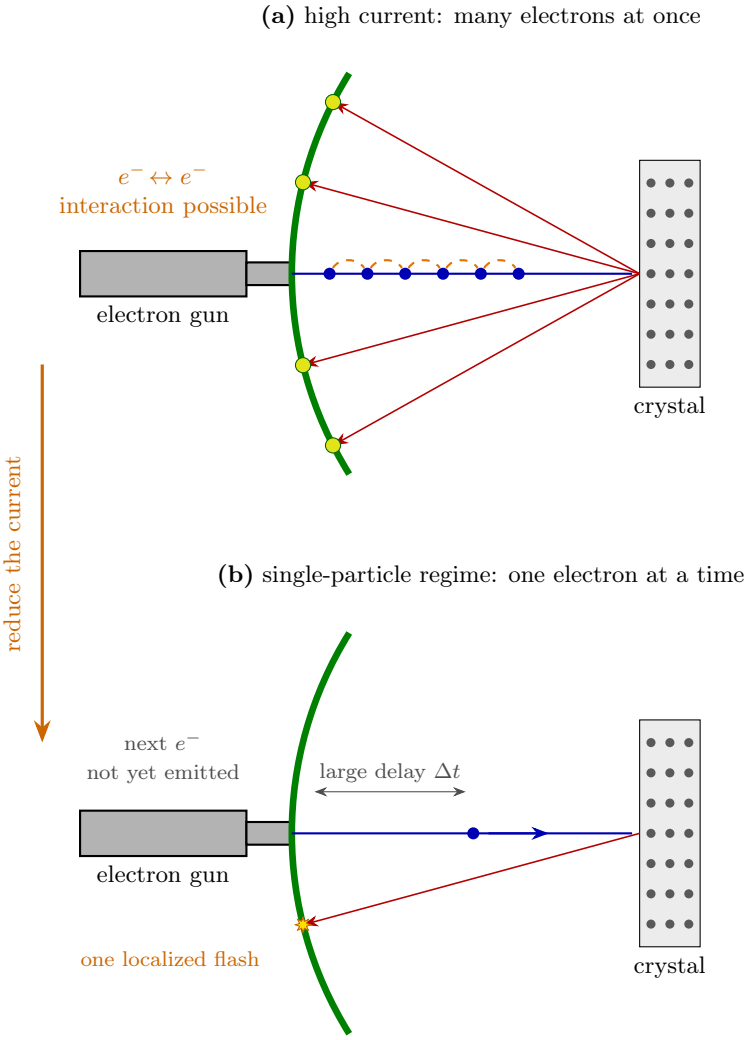


Figure 4: **Single-particle regime.** (a) At high current, many electrons are present simultaneously and can interact with each other. The screen shows a continuous diffraction pattern. (b) At low current, only one electron is present at a time, so electron-electron interaction is excluded. Each electron produces a localized flash on the screen

Why is this experiment important? Because now two different electrons are **not present** in the apparatus at the **same time**. Therefore, they cannot interact with each other during propagation. So if diffraction behaviour is still observed, we can no longer explain it by saying: "*electron 1 interacts with electron 2*" and this collective interaction generates the pattern. That **possibility has been removed**. Now the logic becomes:

many electrons simultaneously	$\Rightarrow$ electron-electron interaction is possible,
one electron at a time	$\Rightarrow$ electron-electron interaction is excluded

So the experiment becomes much interesting.

◎ What is actually detected? Each electron reaches the fluorescent screen individually. When an electron hits the screen, it produces a localized luminous event: the electron excites the fluorescent material and a small flash of light is observed. So experimentally we do not see a continuous faint wave spread over the whole screen. We see:

one electron → one localized detection event

Then some time later:

next electron → another localized detection event

That is already very interesting, because the propagation had shown wave-like diffraction, while the detection is localized. In other words, we have demonstrated that **the electron is detected as a localized event** and, because we send electrons one at a time, we have removed the possibility that the pattern comes from electron-electron interactions.

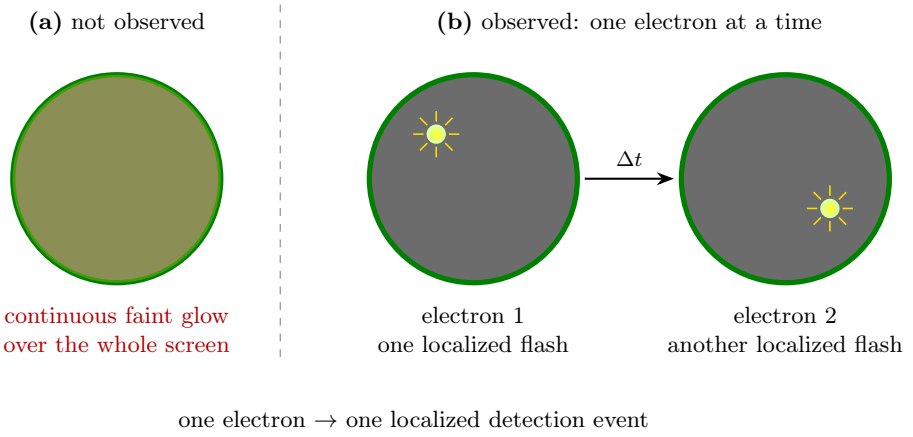


Figure 5: **Detection of individual electrons.** Each electron produces a localized flash on the fluorescent screen. The flashes are separated in time, and after many repetitions they accumulate to form the same diffraction pattern as in Figure 3.

Now, let's **repeat the experiment many times** to see what happens and if the position of the next spot can be predicted in advance.

Localized detection events and the emergence of a diffraction pattern

In the previous figures, electrons are sent **one at a time**. Each electron hits the fluorescent screen and produces one localized flash. So for a single electron, we do not see a little wave spread across the screen, we see one point.

The location of the next flash cannot be predicted from the previous ones. Because successive electrons arrive at different spots, then the next position cannot be determined beforehand. But the main point is that random does not mean **uniformly** random. The flashes do not populate every point of the screen equally. Instead, some regions receive many detections, while other regions receive very few or none.

When many single-electron detections are **accumulated**, those frequencies reproduce the same diffraction structure seen with the ordinary electron beam. In Figure 6, we show a schematic of the accumulation process. The first panel shows only a few detections, which appear scattered and do not reveal any structure. As more and more electrons are sent, the **flashes accumulate**.

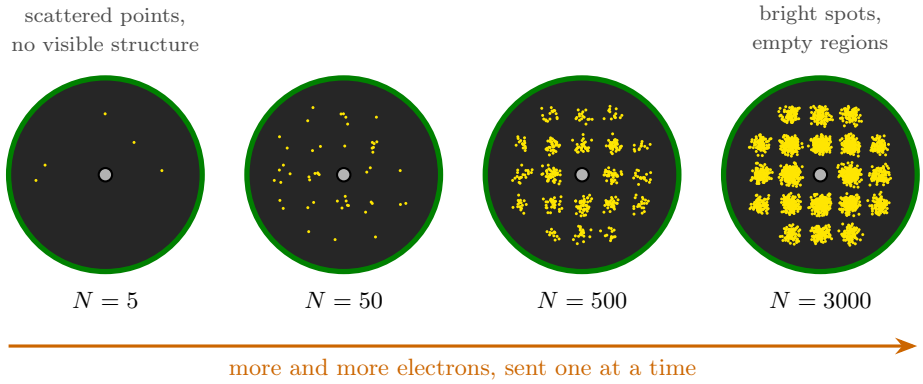


Figure 6: **Accumulation of many single-electron detection events.** Each electron produces a localized flash on the fluorescent screen. After many repetitions, the flashes accumulate to form the same diffraction pattern as in Figure 3.

So after a small number of electrons, we might see something apparently messy ($N = 50$). After more electrons ($N = 500$), and finally after many electrons ($N = 3000$), the diffraction pattern emerges statically.

Why this is surprising. Each electron is detected as a particle:

$$\text{one electron} \rightarrow \text{one localized spot}$$

But the ensemble of many electrons behaves as if some wave-like interference rule were controlling where detections are more or less likely:

$$\text{many individual electrons} \rightarrow \text{diffraction pattern}$$

So we have two apparently contradictory features in the same experiment:

1. At **detection** (one electron at a time), the electron behaves like a **particle**. Because the electron appears at one precise location, not spread over the whole screen.
2. In the **accumulated distribution** (many electrons), the electron behaves like a **wave**. Because the spots organize according to a diffraction pattern.

So if we repeat the experiment a huge number of times, we can say something statistical: "*some positions will be hit much more often than others*". So quantum mechanics is already forcing us toward two levels of description:

- Individual event: **unpredictable** and **localized**.
- Many events: **highly structured statistics**.

Key Takeaways 1

We began by asking how to identify wave behaviour:

$$\text{interference/diffraction} \Rightarrow \text{wave-like behaviour}$$

Then we observed electron diffraction:

$$\text{electrons} \rightarrow \text{diffraction}$$

Then we sent electrons one by one, eliminating electron-electron interactions:

$$\text{one electron at a time}$$

Each electron was detected locally:

$$\text{one electron} \rightarrow \text{one spot}$$

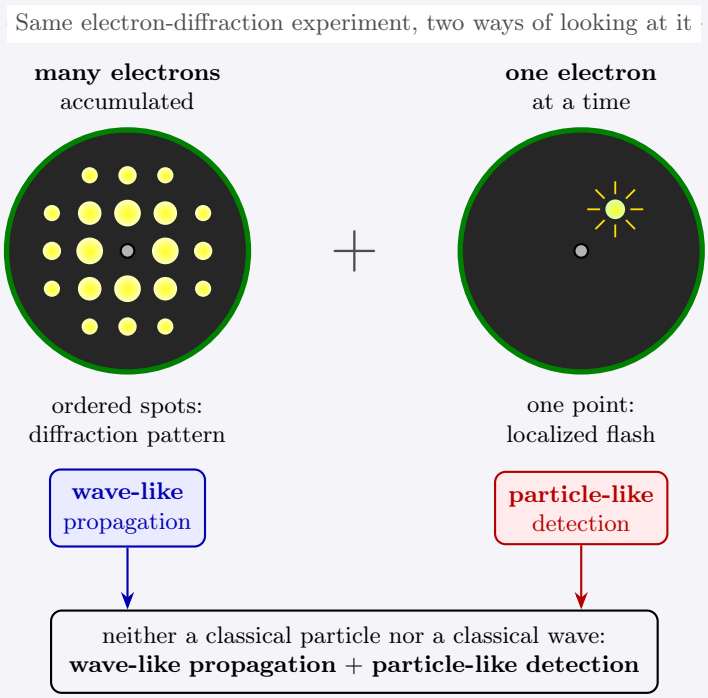
But many such detections reconstructed the diffraction pattern:

$$\text{many individual detections} \rightarrow \text{wave-like diffraction pattern}$$

Therefore, quantum objects exhibit wave-like propagation and particle-like detection. Or even more compactly:

$$\text{interference during propagation} + \text{localization at detection}$$

This is the experimental problem that quantum mechanics now has to explain. And the natural next question is no longer "*wave or particle?*" but: ***why can we predict the distribution but not the individual detection?***


Same electron-diffraction experiment, two ways of looking at it
many electrons
accumulated
one electron
at a time
ordered spots:
diffraction pattern
one point:
localized flash
wave-like
propagation
particle-like
detection
neither a classical particle nor a classical wave:
wave-like propagation + particle-like detection

1.2 Intrinsic Randomness of Quantum Measurements

Individual detection events are unpredictable

We have already established that electrons exhibit **wave-like diffraction** but are detected as **individual, localized particles**. Now we want to investigate a different question: ***can we predict where the next electron will be detected?*** This question introduces one of the most important features of quantum mechanics: **randomness in measurement outcomes**.

Let's consider the single-electron experiment again (page 14):

$$\text{electron gun} \rightarrow \text{crystal} \rightarrow \text{fluorescent screen}$$

We send electrons one at a time, keeping the experimental conditions unchanged. In particular, the electrons are prepared with the same nominal energy and momentum, and the crystal and screen remain in the same positions.

As we have seen, if we repeat the experiment only a few times (e.g., 3 times), we might observe:

$$e_1 \rightarrow x_1, \quad e_2 \rightarrow x_2, \quad e_3 \rightarrow x_3$$

Where e_i is the i-th electron and x_i is the position on the screen where it is detected. Even though we repeat the same preparation procedure, the detection positions x_1, x_2, and x_3 are **different** and **unpredictable**. So the important observation was that the next detection position cannot be determined in advance from the information supplied by the experimental preparation or from the previous detections.

Why was this observation surprising? Because a classical particle, like a bullet, would always hit the same spot on the screen if it is prepared in the same way.

In classical mechanics, if we know a particle's initial position and momentum, together with the forces acting on it, we can in principle calculate its trajectory:

$$(x(0), p(0)) + \text{equations of motion} \longrightarrow x(t)$$

Therefore, we can predict where the particle will arrive. If our classical prediction is inaccurate, we might suspect that we did not know the initial conditions precisely enough.

Instead, quantum mechanics changes this situation. Even when we prepare an electron in the same quantum state and repeat the same measurement, the standard quantum description generally does not provide a definite position for the next detection. Instead, it provides probabilities for different possible outcomes.

At this stage, we do not need to calculate those probabilities. What matters is the experimental observation:

$$\text{same preparation procedure} \not\Rightarrow \text{same detection position}$$

Notice that this does **not** mean quantum mechanics cannot predict anything. It means that **predicting the individual result** and **predicting the statistics of many results** are different tasks.

Does unpredictable mean equally likely everywhere? No. Suppose the fluorescent screen has three possible detection regions: A, B, and C. The electron might arrive in any of them, but it does not follow that

$$P(A) = P(B) = P(C)$$

Some regions may be much more likely than others, and some may have a negligible probability of detection. That is perfectly compatible with the diffraction pattern we observed in the previous section.

unpredictable individual outcome $\neq$ uniformly random outcome

Have we already demonstrated intrinsic randomness? [4] We should distinguish the observation from its interpretation:

- What we have observed is that individual detection positions are unpredictable under our controlled experimental conditions.
- That observation alone does not prove that randomness is fundamentally intrinsic rather than a consequence of unknown variables.

One might still ask whether some inaccessible information determines each outcome, just as an apparently random coin toss can be determined by its initial conditions. We will not answer it yet, but we will investigate it in the next pages.

Now, we have identified the randomness of individual detections. The next question is: ***if individual outcomes are unpredictable, how can the diffraction pattern be so reproducible?***

Key Takeaways 2: Individual detection events are unpredictable

One electron is sent toward the crystal.

$\Downarrow$

One localized detection occurs.

$\Downarrow$

Its position cannot generally be predicted in advance.

Quantum measurements can produce different individual outcomes even when the system is prepared in the same way.

[4] **Intrinsic** means inherent to the nature of something, rather than caused by external factors. In quantum physics, *intrinsic randomness* refers to randomness that is fundamental to the measurement outcome, not merely due to incomplete knowledge or imperfect measuring instruments.

Statistical regularity from many repetitions

Thus we cannot predict where the next electron will be detected, but this raises an apparent contradiction: if individual electron detections are unpredictable, ***how can the diffraction pattern be so regular and reproducible?*** In other words, how can we reconcile the unpredictability of individual events with the predictability of the overall pattern?

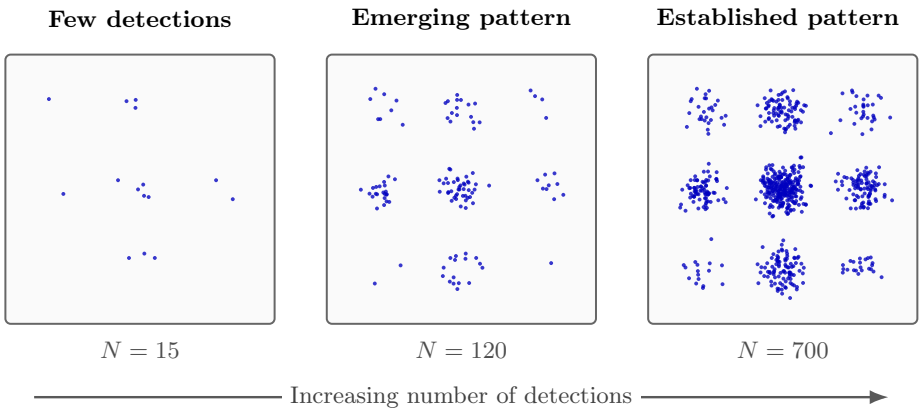
Random individual events do not imply a random overall distribution. Let's return to our electron-diffraction experiment. We prepare the electron gun, crystal and fluorescent screen, and send electrons **one at a time**. For each electron, we observe one localized flash at a position that we cannot predict beforehand.

Now suppose we repeat the experiment N times, without changing the experimental conditions. We observe:

$$x_1, x_2, x_3, \ldots, x_N$$

Where x_i is the detection position of electron i. At first, with only a few electrons, the spots appear scattered without an obvious pattern. However, as the number of detections increases, something remarkable happens: the detections accumulate preferentially in certain regions, reproducing the characteristic diffraction pattern.

This is the same experimental picture developed earlier on page 19, where individual localized flashes appear at unpredictable positions, but they occur with a spatial distribution associated with diffraction.



So we must distinguish between two different types of predictions:

- **Individual prediction**: *where will the next electron arrive?*
- **Statistical prediction**: *how frequently will electrons arrive in different regions after many repetitions?*

Quantum mechanics answers these questions differently.

From individual positions to relative frequencies. Imagine dividing the fluorescent screen into regions A, B, and C. After sending N electrons, we count how many were detected in each region: N_A, N_B, and N_C. If these are all the possible detection regions, then:

$$N_A + N_B + N_C = N$$

We can define the **relative frequency** of detections in region A:

$$f_A(N) = \frac{N_A}{N}$$

For example, imagine that after 1,000 detected electrons we have:

Region	Detections	Relative Frequency
A	700	70%
B	250	25%
C	50	5%

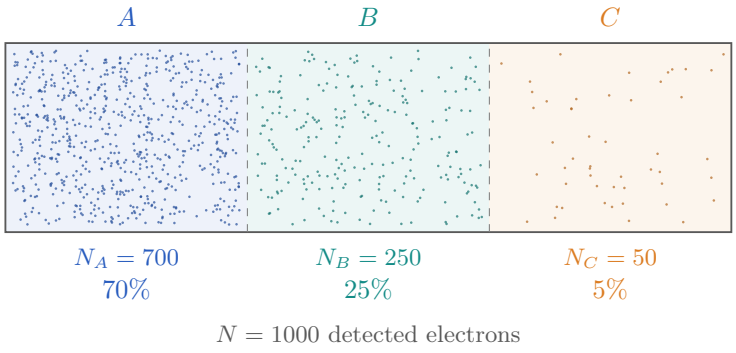


Figure 7: **Relative frequencies of electron detections.** A fluorescent screen is divided into three equal-sized regions A, B, and C. After $N = 1000$ detections, the hypothetical counts are $N_A = 700$, $N_B = 250$, and $N_C = 50$, corresponding to relative frequencies of 70%, 25%, and 5%. Each dot represents one localized electron detection. All values are illustrative.

The **next electron remains unpredictable**, it might arrive in A, B, or C. But if we repeat the same experiment a sufficiently large number of times, the relative frequencies tend to stabilize near characteristic values. For region A:

$$f_A(N) \longrightarrow P(A) \quad \text{as } N \to \infty$$

Under repeated, identically prepared independent trials, $P(A)$ is the probability of obtaining a detection in region A. This does **not** mean that every group of 100 electrons must contain exactly 70 detections in A. The counts still fluctuate (see Figure 8, page 26).

It means that, as the number of repetitions becomes large, the relative fluctuations become less significant and the estimated probabilities become increasingly reliable. This is what we mean by **Statistical Regularity**.

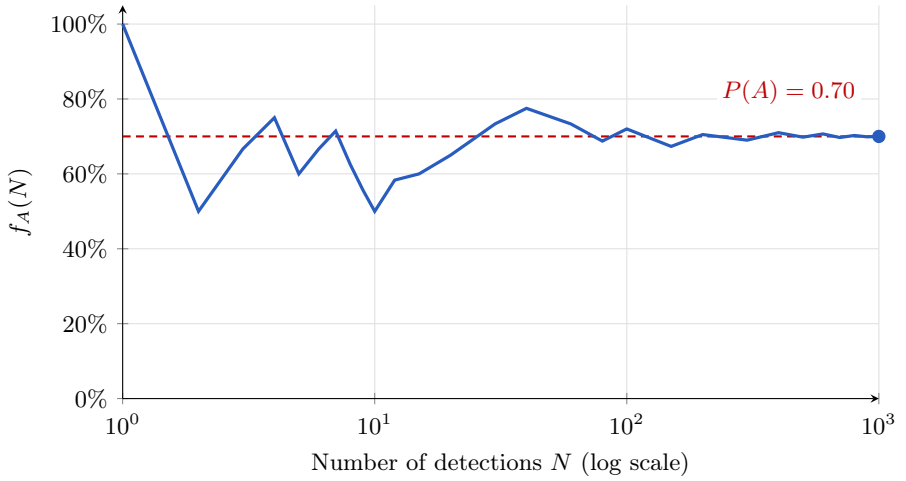


Figure 8: **Statistical convergence of relative frequencies.** The cumulative relative frequency $f_A(N) = N_A/N$ fluctuates as electrons are detected, but tends to stabilize near the hypothetical probability $P(A) = 0.70$ as N increases. The logarithmic horizontal axis emphasizes the fluctuations at small sample sizes. The blue curve connects selected cumulative frequencies, while the dashed red line represents the underlying probability. All values are illustrative.

How does this explain the diffraction pattern? A diffraction pattern contains bright and dark regions. On page 10 we saw that, for classical light, brightness is connected to intensity. So now we can give an experimental interpretation to the electron-diffraction pattern:

- At a position where the electron signal is strong, we observe many detections over time.

  bright diffraction region $\Longrightarrow$ **high** detection **frequency**

- At a position where the signal is weak, we observe fewer detections.

  dark diffraction region $\Longrightarrow$ **low** detection **frequency**

For an ideal detector, the number of electrons detected per unit area and time gives the local electron arrival flux. Thus, the diffraction pattern tells us where electron detections are more or less likely. The pattern is not visible because each electron deposits a fraction of itself in every bright region. It emerges because many complete, individually detected electrons accumulate with different frequencies in different regions.

One electron $\rightarrow$ one localized detection

Many repetitions $\rightarrow$ reproducible spatial distribution

What does quantum mechanics predict? In classical mechanics, we often try to predict the trajectory of each particle, given sufficiently precise initial conditions.

In quantum mechanics, the standard theory generally predicts **probabilities for measurement outcomes**. For example, instead of determining the exact position x at which the next electron will be detected, the theory can predict a probability distribution over possible positions.

Those probabilities can then be tested experimentally by repeating the preparation and measurement many times. Later, we will learn the mathematical rule that connects quantum states with measurement probabilities: the **Born rule**. For now, we do not need that rule. The experimental observation already tells us that our model must explain both unpredictable individual detections and reproducible statistical distributions.

Randomness and regularity are compatible. A random process can follow a well-defined probability distribution. So, randomness describes the uncertainty of individual outcomes, while statistical regularity describes the reproducibility of their distribution.

Individual measurement	Many repeated measurements
One localized flash	A collection of localized flashes
Exact position unpredictable	Relative frequencies become stable
Different outcomes are possible	The overall diffraction pattern is reproducible
Random individual result	Predictable statistical distribution

Key Takeaways 3: Statistical regularity from many repetitions

Individual electron detections are unpredictable
⇓
Repeat the experiment many times
⇓
Relative detection frequencies stabilize
⇓
A reproducible diffraction pattern emerges

Quantum Mechanics does not need to predict each individual detection to make precise experimental predictions. It must correctly predict the statistical distribution of possible outcomes.

Why the diffraction pattern is not caused by interactions between particles

We have already encountered this topic on page 16, so this part is mainly about understanding ***why the single-particle experiment is such an important experimental test***.

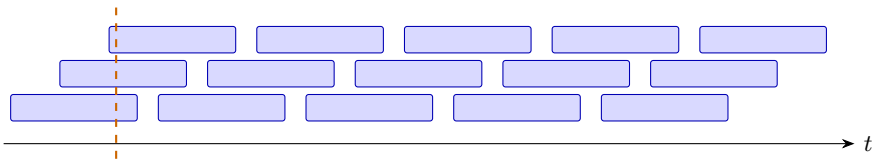
Our initial observation was that an electron beam produces a diffraction pattern. We interpreted this as evidence of wave-like behaviour. But there is a possible objection: ***what if the diffraction pattern is caused by interactions between electrons rather than by the wave-like nature of electrons?***

This is not an unreasonable hypothesis. Electrons have negative electric charge. Consequently, they exert repulsive electric forces on one another. If many electrons travel through the apparatus[5] simultaneously, they may influence each other's motion. One could therefore hypothesize that their collective interactions produce the observed spatial distribution. Thus, the question becomes: ***how can we test this hypothesis experimentally?***

Eliminate electron-electron interactions. We change a simple experimental parameter: the **electron emission rate**. We reduce the beam current until successive electrons are sufficiently separated in time that only one incident electron is inside the apparatus at any given moment.

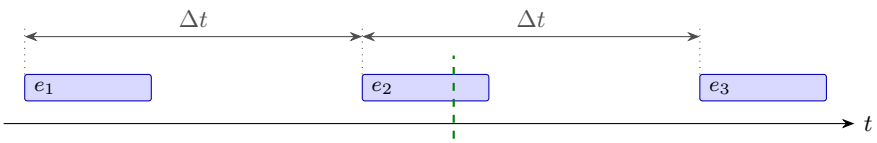
Mathematically, if t_f is the time needed for an electron to travel through the apparatus, we want the interval between emissions to satisfy approximately: $\Delta t \gg t_f$.

(a) high rate: $\Delta t \lesssim t_f$



3 electrons inside at once: possible interaction

(b) low rate: $\Delta t \gg t_f$



only 1 electron inside:
no $e^- \leftrightarrow e^-$ interaction

[5] The apparatus is the complete experimental setup used to perform electron diffraction: the electron gun, the crystal, and the fluorescent screen. The electron gun emits electrons, the crystal scatters them, and the fluorescent screen detects their arrival (see page 17).

Now electron e_1 has completed its journey before electron e_2 enters the apparatus. Therefore, the two incident electrons **cannot interact** with one another during their passage.

What happens experimentally? We already know the answer:

- First, each electron produces one localized flash on the fluorescent screen.
- Then, after repeating the experiment many times, the accumulated detections reproduce the diffraction pattern.

This result is decisive for our hypothesis, because if the diffraction pattern required interactions **between incident electrons**, it should disappear when those interactions are eliminated. But it does **not disappear**. Therefore, electron-electron interactions are not necessary for diffraction.

Does this mean the electron does not interact with anything? No! We have eliminated interactions **between different incident electrons**, but the incident electron still interacts with the crystal. The electron is scattered by the periodic structure of the crystal. Without the scattering structure, this particular diffraction experiment would **not produce the same pattern**.

$$\underbrace{\text{electron} \leftrightarrow \text{electron}}_{\text{excluded by the experiment}} \qquad \underbrace{\text{electron} \leftrightarrow \text{crystal}}_{\text{still present (and necessary?)}}$$

Key Takeaways 4: Why the diffraction pattern is not caused by interactions between particles

Hypothesis: diffraction is caused by
interactions among incident electrons
$\Downarrow$
Send one electron at a time
$\Downarrow$
Electron-electron interactions are eliminated
$\Downarrow$
The diffraction pattern still emerges
$\Downarrow$
The hypothesis is rejected

Wave-like diffraction is not a collective effect produced by interactions among the incident electrons. It is a feature that our description of individual electrons must accommodate (i.e., *must be capable of explaining the phenomenon*), or in other words, our mathematical model of a single electron must be able to explain why repeated single-electron experiments produce a diffraction pattern.

In summary, we need a theory describing even a single electron must contain the ingredients necessary to predict the diffraction pattern.

⚖ Quantum randomness vs. ordinary classical ignorance

We have established two important experimental facts:

- **Individual detections are unpredictable**: we cannot determine where the next electron will arrive.
- **Statistical outcomes are reproducible**: after many repetitions, the detections form a predictable diffraction pattern.

But there is a fundamental question we have not answered yet: ***is the outcome genuinely random, or do we simply lack information?*** To understand this distinction, let's start with something familiar.

Classical randomness: tossing a coin. Imagine tossing a fair coin. Before the coin lands, we typically assign:

$$P(H) = \frac{1}{2}, \qquad P(T) = \frac{1}{2}$$

Where H and T denote heads and tails, respectively. We call the **result random** because we cannot reliably predict whether it will be heads or tails. But *why can't we predict it?* Because we do not know every detail of the coin's motion: its initial velocity, angular velocity, orientation, aerodynamic forces, and so on. In classical mechanics, these quantities determine its trajectory.

Therefore, if we knew the complete initial conditions and could solve the equations of motion exactly, we could, in principle, predict the result. The randomness arises from **our incomplete knowledge**, not necessarily from an intrinsic randomness in the physical laws. This is often called **Epistemic Randomness**.[6]

Classical randomness = uncertainty due to incomplete information

In contrast with **Ontic Randomness** (or *intrinsic randomness*), where unpredictability is considered part of the physical process itself, rather than merely a limitation of our knowledge.

Quantum randomness: a single photon at a beam splitter. Now consider the example where a beam splitter (i.e., a partially reflective mirror) is an optical component that can transmit or reflect light (we will examine the experiment in the next sections, see Figure 9, page 31). For an ideal 50/50 beam splitter, a single photon entering through a specified input has two possible detection outcomes.

We can describe the outcomes as:

$$P(D_1) = \frac{1}{2}, \qquad P(D_2) = \frac{1}{2}$$

Where D_1 and D_2 denote the two possible detectors. Each photon produces a detection in one output, but we cannot predict which one. At first glance, this looks exactly like tossing a coin.

[6] **Epistemic** means *related to knowledge* (from the Greek epistēmē, meaning knowledge). So **epistemic randomness** is randomness caused by **our lack of information**, rather than by something fundamentally unpredictable in nature.

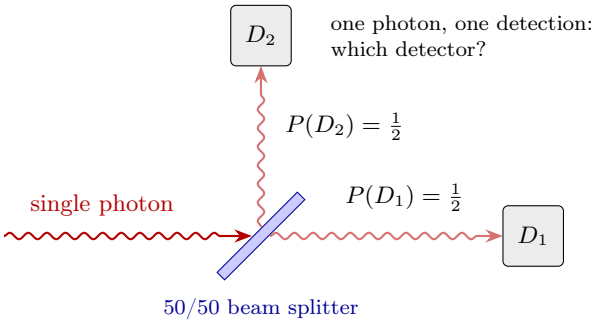


Figure 9: Conceptual 50/50 beam splitter.

⚖ **But is it really the same as tossing a coin?** Suppose we make the following hypothesis: "*perhaps the photon follows a definite path determined by some microscopic properties that we simply do not know*".

For instance, maybe a more detailed description of what happens inside the beam splitter would reveal the missing information. This would make quantum unpredictability analogous to classical ignorance.

Classical ignorance	**Intrinsic quantum randomness**
Outcome appears random because information is missing	Outcome is probabilistic even with a complete quantum-state description
Complete classical initial conditions determine the result	The standard quantum theory assigns probabilities to possible results
Example: classical coin toss	Example: single photon at a beam splitter

In the **standard formulation of quantum mechanics**, an exact description of the prepared quantum state generally allows us to calculate measurement probabilities, but not to determine which individual outcome will occur. Thus:

$$\text{Complete quantum state} \longrightarrow \text{probabilities of measurement outcomes}$$

Rather than, in general:

$$\text{Complete quantum state} \not\longrightarrow \text{one predetermined measurement outcome}$$

🔍 **Can we already prove that quantum randomness is intrinsic?** No, not from the experiments we have discussed so far. Observing a 50/50 distribution does not, by itself, prove fundamental randomness. A classical coin can also produce a 50/50 distribution. So the question is not simply whether the outcomes look random,but whether a classical explanation based on unknown, predetermined outcomes can reproduce all the experimental observations.

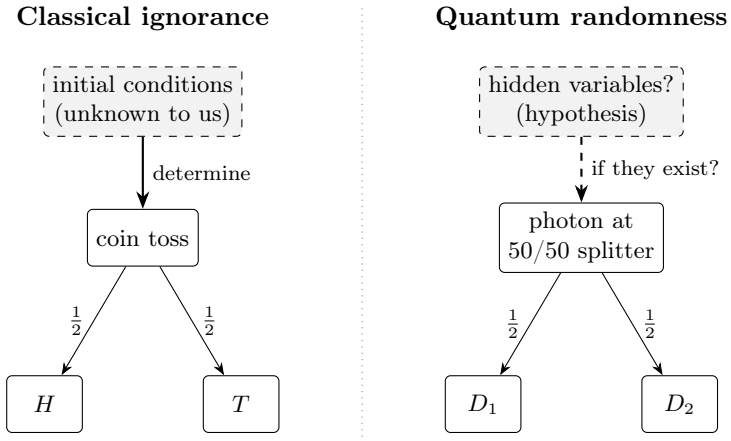


Figure 10: Both situations give a 50/50 distribution, but only in the classical case do we know the randomness comes from missing information.

Key Takeaways 5: Quantum randomness vs. ordinary classical ignorance

Classical randomness:	we don't know enough.
Quantum randomness:	the standard theory predicts probabilities, not individual outcomes.

The difference is not that classical physics never uses probability, it does! The **difference** is what probability represents: incomplete information about a classically determined event (classical), versus a fundamental part of the standard quantum description of measurement outcomes (quantum).

Key Takeaways 6

Individual randomness, statistical predictability.
When electrons are sent one at a time, each produces a localized detection at an unpredictable position. However, after many repetitions, the detections accumulate into a reproducible diffraction pattern.

One electron → unpredictable detection

Many repetitions → predictable statistical pattern

The pattern persists even when electrons cannot interact with one another. Therefore, electron-electron interactions are not required to explain diffraction.

Is the randomness fundamental? Classical randomness can arise from incomplete knowledge of initial conditions. In the standard formulation of quantum mechanics, even a fully specified quantum state generally predicts probabilities rather than individual outcomes.
However, the experiments discussed so far do not prove that quantum randomness is intrinsic.

Quantum mechanics must explain both the unpredictability of individual detections and the reproducible, wave-like statistical pattern. *How can nature produce unpredictable individual events while following highly reproducible statistical laws?*

1.3 Beam Splitters and the Mach-Zehnder Interferometer

1.3.1 The single beam splitter

Until now, we have mainly studied electrons. We observed that their individual detection positions were unpredictable, even though many repeated detections produced a reproducible diffraction pattern. Now we will investigate the same underlying problem using **photons**, the particles associated with light. Our central question remains: ***what happens when we send just one quantum particle?***

What is a beam splitter? A **Beam Splitter** is an optical component that divides an incoming light beam (i.e., a stream of photons) into two outgoing beams. It can be implemented, for example, using a partially reflective mirror.

For an ideal **50/50 beam splitter**:

- 50% of the incident light intensity is **transmitted**.
- 50% of the incident light intensity is **reflected**.

So far, nothing surprising, because ordinary classical wave optics already explains how light can be divided into two beams. But now let's change the experiment.

What if we send only one photon? Instead of sending an intense beam of light, we consider individual photons entering the beam splitter. We place two detectors at the outputs:

- D_1 measures the **transmitted output**.
- D_2 measures the **reflected output**.

Each detector can produce a click (i.e., a signal) when it detects a photon.

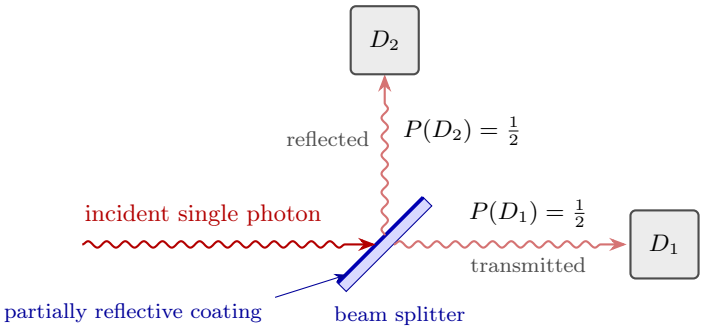
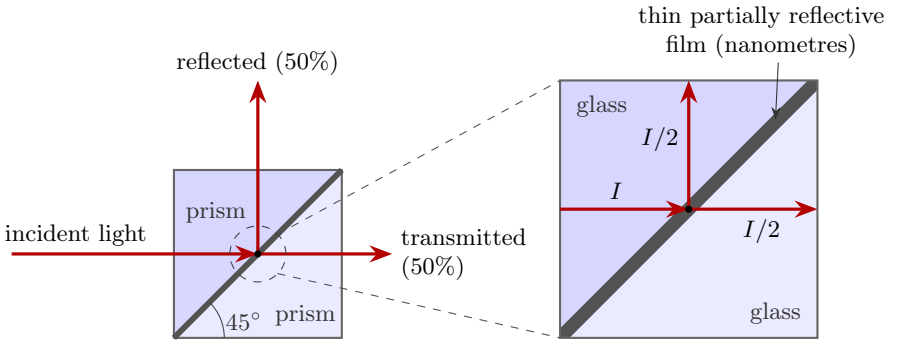


Figure 11: **A single photon at a 50/50 beam splitter.** The two outputs are equally likely detection outcomes, and each photon is detected at exactly one detector.

What do we observe? For an ideal single-photon experiment with perfect detectors, one photon produces **one detection event**: either at D_1 or at D_2.



(a) side view of the cube *(b) close-up of the coating (not to scale)*

Figure 12: **How a beam splitter works.** (a) A beam-splitter cube is made of two glass prisms joined along their diagonal. The light hits the interface at 45°: half of the intensity is transmitted straight on, and half is reflected at 90°. (b) Close-up of the interface: the "partially reflective mirror" is a very thin coating, only a few nanometres thick, which reflects part of the light and lets the rest through. For a 50/50 beam splitter, the coating is tuned so that an incident intensity I leaves as $I/2 + I/2$.

Thus, the probabilities are:

$$P(D_1) = \frac{1}{2}, \qquad P(D_2) = \frac{1}{2}$$

However, notice the difference between two statements:

- **With a classical light beam**, 50/50 refers to how the incident intensity is distributed between the outputs.
- **With a single photon**, 50/50 refers to the probabilities of the two possible detection outcomes.

We do **not** detect half a photon at D_1 and half a photon at D_2. So individual photon detections occur at one detector rather than being divided into fractional detection events.

Deepening: Particle indivisibility in detection

Classic light beam. Imagine sending a continuous light beam with intensity I toward a 50/50 beam splitter. The two outputs have intensities:

$$I_1 = \frac{I}{2}, \qquad I_2 = \frac{I}{2}$$

Both detectors receive lights simultaneously.

Single photon. Now imagine sending **exactly one photon** toward the beam splitter. There are two possible detection outcomes:

- D_1 clicks and D_2 does not.
- D_2 clicks and D_1 does not.

For an ideal experiment, each outcome has a probability of 50%. What we do **not** observe is a single photon depositing half its energy at each detector. If the photon's energy is $E = h\nu$, then an ideal detection registers the photon's entire energy:

$$\begin{aligned} & D_1 : E, \qquad D_2 : 0 \\ \text{or} \quad & D_1 : 0, \qquad D_2 : E \end{aligned}$$

Not:

$$D_1 : \frac{E}{2}, \qquad D_2 : \frac{E}{2}$$

This is what we mean by **particle indivisibility in detection**.

But this does not mean the photon necessarily travels along one definite classical path. We are describing what happens *when it is detected*, not asserting what its trajectory was before detection.

What if we repeat the experiment? Imagine sending $1'000$ photons, one at a time. A possible sequence of detector outcomes is:

$$D_1,\ D_2,\ D_2,\ D_1,\ D_1,\ D_2, \ldots$$

We **cannot** generally predict which detector will register the next photon. But after sufficiently many repetitions, approximately half of the photons will be detected at each output:

$$\frac{N_1}{N} \approx 0.5, \qquad \frac{N_2}{N} \approx 0.5$$

This should sound familiar! It is exactly the distinction we established in previous sections:

$$\text{unpredictable individual detections} + \text{predictable statistics}$$

Anticorrelated detector clicks and particle indivisibility

Let's examine the detection events more closely. Imagine sending photons into the beam splitter one at a time. We observe the following possible sequence:

Photon	**Detector** D_1	**Detector** D_2
1	✔ Click	–
2	–	✔ Click
3	–	✔ Click
4	✔ Click	–
5	✔ Click	–

Whenever D_1 detects the photon, D_2 does not detect that same photon, and vice versa. This relationship is called **Anticorrelation**. In other words, detecting the photon at one output **excludes** detecting the **same photon** at the other output.

$$D_1 \text{ clicks} \implies D_2 \text{ does not click}$$
$$D_2 \text{ clicks} \implies D_1 \text{ does not click}$$

This is what we mean by "*anticorrelated detector clicks*".

Why is this observation important? When a continuous light beam enters, the outgoing intensity is distributed equally:

$$I_1 = \frac{I}{2}, \qquad I_2 = \frac{I}{2}$$

Both outputs carry light. Now consider a single photon. If the photon behaved like a **classical wave** whose energy was simply **divided into two equal portions**, we might expect half its energy to be detected at each output. But that's not what we observe! Instead, the photon produces a localized detection at one output or the other.

$$P(D_1 \text{ only}) = \frac{1}{2}, \qquad P(D_2 \text{ only}) = \frac{1}{2}, \qquad P(D_1 \text{ and } D_2) = 0$$

The third expression says that a single photon does not produce simultaneous detection events at both detectors.

Particle indivisibility. Suppose a photon has energy:

$$E = h\nu$$

The two possible ideal detection outcomes are:

$$(E_1, E_2) = \begin{cases} (h\nu, 0) & \text{photon detected at } D_1 \\ (0, h\nu) & \text{photon detected at } D_2 \end{cases}$$

We do not observe the single photon's energy split into two separately detected halves. So **the photon is indivisible in detection**. But there is a subtle distinction worth preserving: anticorrelated clicks tell us **where the photon is detected**, not necessarily which definite classical trajectory it followed **before** detection. Therefore, we must **not conclude** that the photon secretly chose one path before reaching the beam splitter. That is a possible classical-style hypothesis, not something established by anticorrelation alone.

Anticorrelation does not contradict the 50/50 probabilities. Let's be precise about the difference between the two statements:

- A **50/50 probability distribution** means that, after many repetitions, approximately half the photons are detected at D_1, $N_1 \approx \frac{N}{2}$, and half at D_2, $N_2 \approx \frac{N}{2}$.
- **Anticorrelation** means that for each individual photon, the two ideal detector outcomes are mutually exclusive (never both occur).

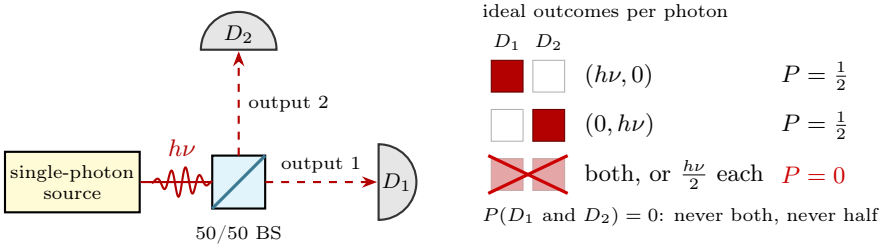
We can express this with conditional probabilities:

$$P(D_2 \text{ clicks} \mid D_1 \text{ clicks}) = 0, \qquad P(D_1 \text{ clicks} \mid D_2 \text{ clicks}) = 0$$

Considering the first conditional probability, this means that given that D_1 detected this photon, the probability that D_2 also detected the same photon is zero. These two facts are completely compatible:

- Over many photons, both detectors register approximately equal numbers of clicks (50/50 statistics).
- For each individual photon, exactly one detector clicks (anticorrelation).

(a) One photon at a 50/50 beam splitter



(b) Click record, one photon at a time

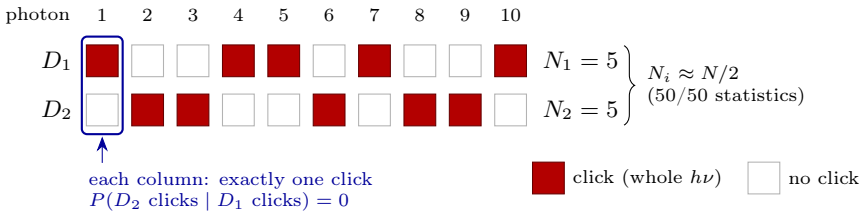


Figure 13: Anticorrelated detector clicks at a 50/50 beam splitter. (a) A single photon of energy $h\nu$ is detected whole, at D_1 or at D_2, never at both and never split: $P(D_1 \text{ and } D_2) = 0$. Dashed lines mark the two possible output ports, not a trajectory the photon is known to follow. (b) An illustrative sequence of detections. Read by columns, the record shows anticorrelation (one click per photon); read by rows, it shows the 50/50 statistics $N_1 \approx N_2 \approx N/2$ (real runs fluctuate around $N/2$).

Why a beam splitter resembles a random 50/50 process

We have established two important facts about a single photon entering an ideal 50/50 beam splitter:

1. **Unpredictability**: we cannot predict which detector will register the photon.
2. **Anticorrelation**: only one detector registers that photon, never both simultaneously.

But now we have a natural question: ***is a beam splitter simply tossing a coin for each photon?***

The analogy with tossing a coin. Imagine an ordinary fair coin. Each toss produces one of two possible outcomes:

$$P(H) = \frac{1}{2}, \qquad P(T) = \frac{1}{2}$$

Similarly, a photon entering a 50/50 beam splitter produces one of two possible detection outcomes:

$$P(D_1) = \frac{1}{2}, \qquad P(D_2) = \frac{1}{2}$$

If we repeat either experiment 1′000 times, we expect approximately 500 occurrences of each outcome. From the perspective of **individual results and their frequencies**, the two experiments look remarkably similar. But does that mean they are governed by the same kind of randomness?

Could the photon behave like a classical coin? In classical mechanics, a coin toss appears random because we do not know the exact initial conditions (remark page 30). *What if the photon behaves similarly?* Perhaps the photon has some unknown microscopic properties that determine whether it will ultimately be detected at D_1 or D_2.

For example, we could hypothesize:

Photon enters the beam splitter
$\Downarrow$
Some unknown physical details determine the outcome
$\Downarrow$
D_1 or D_2

In this hypothetical picture, the 50/50 statistics arise because we lack information about those microscopic details. This is the **Classical-Ignorance Hypothesis**: an event appears random only because we do not know all the physical information that determines its outcome.

Can the single beam-splitter experiment distinguish the two explanations? Similar to the single-particle regime of the double-slit experiment, a single 50/50 beam splitter cannot, by itself, distinguish quantum probabilities

from a classical coin-toss model. All we have established is that its outcomes **behave statistically like a fair coin toss**. Whether the underlying physics is the same remains an open question.

How can we investigate the difference? We need an experiment in which the classical random-choice model and quantum mechanics make **different predictions**.

Instead of studying one beam splitter in isolation, what if we connect two beam splitters using mirrors, allowing the two possible routes to meet again? This leads us to the **Mach-Zehnder interferometer**. The idea is really simple: if each beam splitter acts like an independent coin toss, combining two of them should still produce the output probabilities predicted by those random choices. In other words, if each beam splitter behaves like a fair coin toss, then passing a photon through two beam splitters should be equivalent to tossing a coin twice.

Key Takeaways 7

A single photon at a 50/50 beam splitter.
An ideal 50/50 beam splitter produces two equally probable detection outcomes:

$$P(D_1) = P(D_2) = \frac{1}{2}$$

However, one photon produces one complete detection event, either at D_1 or at D_2, never at both simultaneously (anticorrelation). The photon is **indivisible in detection**.

The classical coin-toss analogy.
The 50/50 statistics resemble tossing a fair coin. Perhaps the outcome is determined by unknown microscopic information, as in classical mechanics (the *classical-ignorance hypothesis*).

However, identical probabilities do not imply identical physical mechanisms. A single beam splitter cannot distinguish between an ordinary classical random-choice model and the quantum description.

The Mach-Zehnder interferometer. If a beam splitter really behaves like a classical coin toss, should two successive beam splitters behave like two independent coin tosses? The Mach-Zehnder interferometer will test this idea.

1.3.2 Construction of the Mach-Zehnder interferometer

We have just discovered that a single photon entering a 50/50 beam splitter behaves, statistically, like a fair coin toss:

$$P(D_1) = P(D_2) = \frac{1}{2}$$

But we still don't know whether this randomness is equivalent to classical ignorance. ***How can we test whether the coin-toss analogy is sufficient?*** We can construct a new experiment using two beam splitters instead of one. This arrangement is called the **Mach-Zehnder Interferometer**.

The experimental setup. The **Mach-Zehnder Interferometer** is an experimental apparatus used to study **interference between two alternative paths** that light (or a quantum particle) can take. It consists of two 50/50 beam splitters, two ordinary mirrors, a single-photon source, and two detectors.

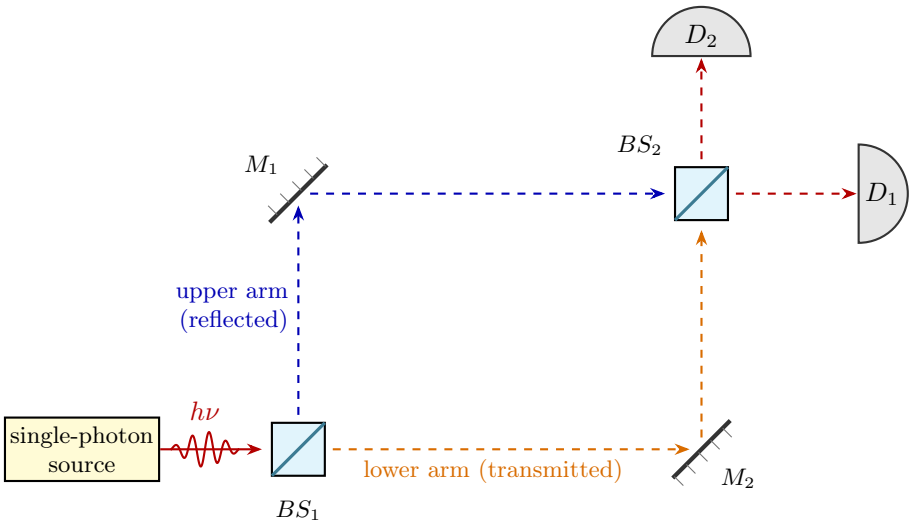


Figure 14: **Mach-Zehnder interferometer.** BS_1 and BS_2 are 50/50 beam splitters; M_1 and M_2 are fully reflecting mirrors. The colored lines show the two available routes, *not* two pieces of a photon; the red dashed lines mark the two output ports leading to the detectors D_1 and D_2.

Let's analyze the Mach-Zehnder interferometer in detail:

- **The first beam splitter BS_1 creates two possible paths for the photon.** A single photon reaches the first beam splitter, BS_1. Just as in the previous section, there are two possible outgoing directions:
  - **Upper arm**: the reflected output.
  - **Lower arm**: the transmitted output.

  We know that, if these paths were separately measured immediately after BS_1, each would have a 50% detection probability. But in this experiment,

we **do not place detectors on the two arms**, instead we allow both possible routes to continue.

⚠ **Remark:** drawing two paths does not mean the photon has been divided into two half-photons. Nor have we established that it secretly chose one definite path. For now, we simply identify the two alternatives available in the apparatus.

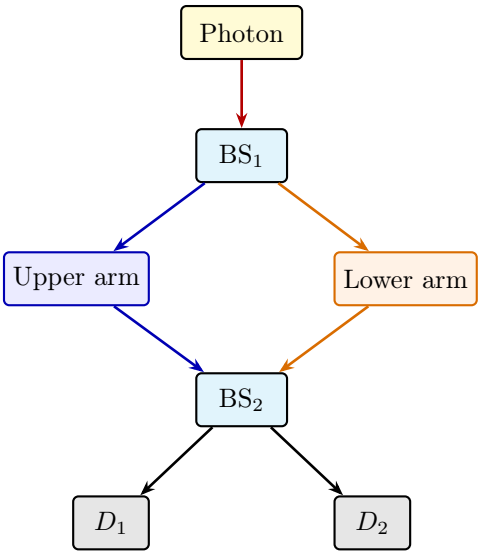
- **The mirrors M_1 and M_2 redirect the two paths.** Next, each path encounters an ordinary mirror:
  - M_1 redirects the upper arm toward the second beam splitter.
  - M_2 redirects the lower arm toward the second beam splitter.

  We assume the mirrors are ideal, with 100% reflectivity. Unlike a beam splitter, **an ordinary mirror does not create two alternative outgoing paths** in this setup; it redirects the incident beam. The goal of the mirrors is simply to bring the two arms together at the second beam splitter, BS_2.

- **The second beam splitter BS_2 brings the alternatives together.** This is the essential new ingredient, because in our previous experiment, we had:

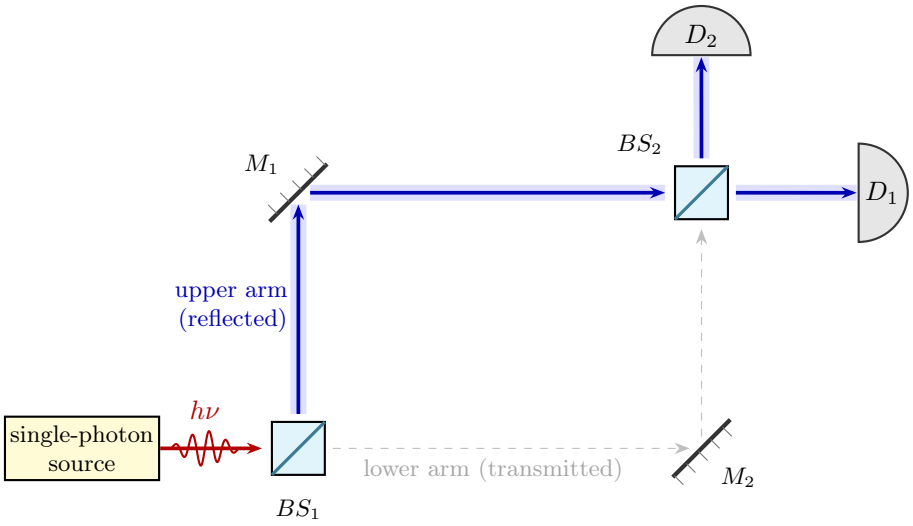
$$\text{Photon} \to BS_1 \to \begin{cases} D_1 \\ D_2 \end{cases}$$

  But now, the photon has two alternative paths that are recombined at BS_2:

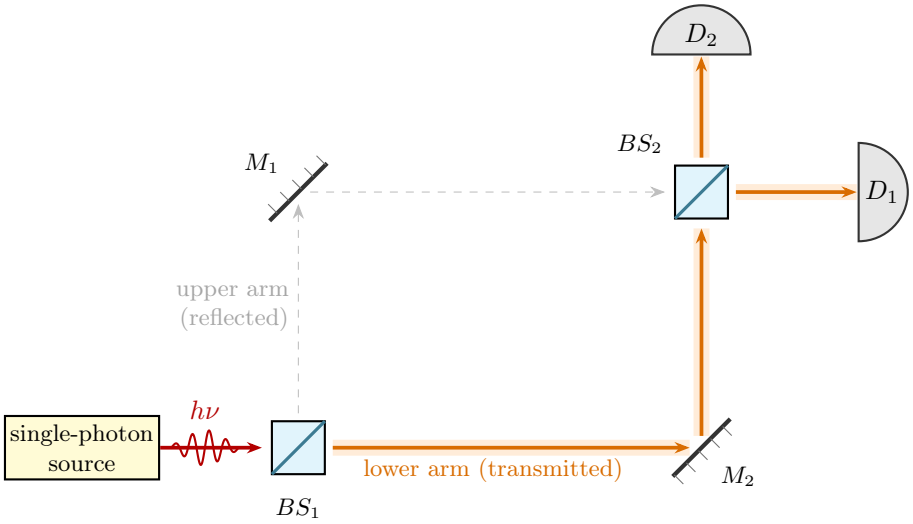


Both routes lead to the **same second beam splitter**, BS_2, and after this point, we place two detectors, D_1 and D_2, to record the outcome. This means that, for a detection at one of the final outputs, the apparatus contains **two possible contributing routes** from the original source:

– **Route 1:** $S \to BS_1 \to M_1 \to BS_2 \to D_1$ or D_2



– **Route 2:** $S \to BS_1 \to M_2 \to BS_2 \to D_1$ or D_2



What are we trying to discover? We now have two beam splitters, each individually characterized as 50/50. If we treat them as ordinary classical devices that make independent random choices, what should the final detectors observe? Would we still expect:

$$P(D_1) = P(D_2) = \frac{1}{2} \; ?$$

Or could bringing the two paths together change the outcome statistics?

Why two independent 50/50 coin tosses give the wrong prediction

We have constructed the Mach-Zehnder interferometer using two 50/50 beam splitters, two mirrors, and two detectors. Now we want to answer a simple question: ***what should we observe if each beam splitter acts like a fair coin toss?***

- **First beam splitter: the first coin toss.** A single photon enters BS_1. Under our classical random-choice hypothesis, the photon randomly chooses one of the two available paths:

$$P(U) = \frac{1}{2}, \qquad P(L) = \frac{1}{2}$$

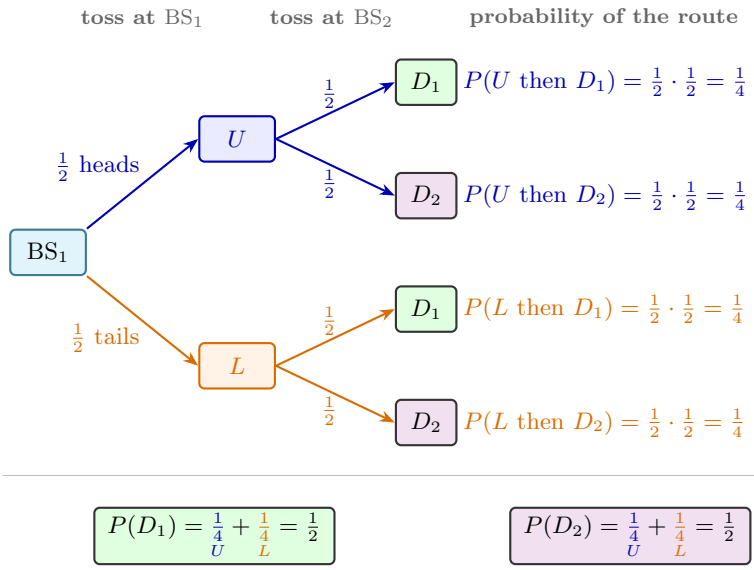
  Where U and L denote the upper and lower arms, respectively. So, according to this model, we imagine that the photon takes **one definite path**, although we do not know which one. This is exactly the coin-toss analogy: heads (upper arm) or tails (lower arm). But remember that this is our **hypothesis**, not a proven fact.

- **Second beam splitter: the second coin toss.** The mirrors redirect the paths toward BS_2. Let's assume the second beam splitter works like another independent 50/50 coin toss. Regardless of which arm the photon came from, it has:

$$P(D_1 \mid U) = P(D_2 \mid U) = \frac{1}{2}, \quad P(D_1 \mid L) = P(D_2 \mid L) = \frac{1}{2}$$

  Where $P(D_1 \mid U)$ means the probability of reaching detector D_1 given that the photon followed the upper arm.

Calculate the four possible outcomes. If the two coin tosses are independent, the probability of a particular sequence is the product of the two probabilities:



$$P(D_1) = \underset{U}{\frac{1}{4}} + \underset{L}{\frac{1}{4}} = \frac{1}{2} \qquad P(D_2) = \underset{U}{\frac{1}{4}} + \underset{L}{\frac{1}{4}} = \frac{1}{2}$$

Notice that **each final detector can be reached through either of the two arms**, so we can sum the probabilities for each detector:

$$P(D_1) = P(U \text{ then } D_1) + P(L \text{ then } D_1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = 50\%$$

$$P(D_2) = P(U \text{ then } D_2) + P(L \text{ then } D_2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = 50\%$$

Even though we have two beam splitters, the final statistics still resemble those of a fair coin toss.

But what happens experimentally? In a properly aligned Mach-Zehnder interferometer, with coherent paths and suitable relative phase, we can observe:

$$P(D_1) = 100\%, \qquad P(D_2) = 0\%$$

All successfully detected photons arrive at the same output! Of course, we can also arrange the interferometer so that the opposite detector receives all the photons. The exact output depends on the relative phase accumulated along the two arms and the beam-splitter configuration. But the important fact is that **the result can be completely different from 50/50**.

Why does the classical prediction fail?

Let's identify exactly what went wrong. Our classical model **assumed** that:

1. The first beam splitter randomly selects one definite path.
2. The second beam splitter makes another independent random choice.
3. We can calculate final probabilities simply by adding the probabilities of different routes.

But the experiment shows that this model is insufficient. The two alternatives cannot always be treated as mutually exclusive classical routes whose probabilities are independently added.

In quantum mechanics, the two paths contribute *probability amplitudes*, and these amplitudes can **interfere** when the paths recombine. That is the missing ingredient. So, the failure of this simple two-coin-toss model does not, by itself, disprove every possible hidden-variable interpretation. It shows that we cannot describe this interferometer as two independent classical random decisions.

And the next question is: ***how can two individually 50/50 beam splitters produce a deterministic output?*** In this case, "*deterministic*" means that all of the photons are detected at one output port, and none are detected at the other.

Interference at the second beam splitter

There is a crucial observation about the Mach-Zehnder interferometer. A photon detected at D_1 has two possible path contributions leading to that outcome: one associated with the upper arm and one with the lower arm. The same is true for a photon detected at D_2. In our classical calculation, we treated those alternatives as independent probabilities. But quantum mechanics requires something different when the alternatives remain coherent.

Probabilities vs. Amplitudes. When two waves meet (page 8), we first add their **amplitudes**. Depending on their relative phase, they can reinforce or cancel each other. Quantum mechanics uses a similar principle: instead of assigning only probabilities to the two paths, we associate them with **probability amplitudes**. These amplitudes are **mathematical quantities that can interfere**; unlike probabilities, they can have **positive**, **negative** or **complex values**.

For a particular detector, suppose the two path contributions have amplitudes a_U and a_L. Then the quantum rule is:

$$P(D) = |a_U + a_L|^2$$

Rather than simply:

$$P(D) = |a_U|^2 + |a_L|^2$$

The second expression is appropriate when the path alternatives contribute incoherently, as in the classical random-path model. Instead, the first expression contains an additional possibility: **interference**.

How can interference change the final probabilities? Similar to the interference of classical waves, the two amplitudes can reinforce or cancel each other:

- If their amplitudes reinforce each other, we obtain **constructive interference**.
- If their amplitudes cancel each other, we obtain **destructive interference**.

For example, consider an ideal, appropriately tuned interferometer in which the contributions are:

Detector	Upper-arm	Lower-arm	Result
D_1	$+\frac{1}{2}$	$+\frac{1}{2}$	Constructive interference
D_2	$+\frac{1}{2}$	$-\frac{1}{2}$	Destructive interference

Then:

$$P(D_1) = \left|\frac{1}{2} + \frac{1}{2}\right|^2 = 1 = 100\%$$

While:

$$P(D_2) = \left|\frac{1}{2} - \frac{1}{2}\right|^2 = 0 = 0\%$$

This example explains how the second beam splitter can produce a deterministic output even though each beam splitter individually has 50/50 transmission and reflection probabilities for a photon arriving through a single input.

Why does this contradict the coin-toss analogy? A classical coin toss has no interference between its possible outcomes.

For example, if we consider two mutually exclusive ways of reaching D_1, classical probability tells us to add their probabilities:

$$P(D_1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

But in the Mach-Zehnder interferometer, the two path contributions **can interfere**, so we must add their amplitudes before calculating the probability. That is the **missing ingredient in our classical model**.

Note (*again*): we are **not** saying the photon physically divides into two smaller photons. Rather, the quantum description of a single photon contains contributions associated with both paths, and these contributions can interfere when the paths recombine.

Equal paths: deterministic output at one detector

At this point, we have established that the two possible paths in a Mach-Zehnder interferometer can interfere when they meet at the second beam splitter. Now let's examine the particular experimental condition that produces the surprising result:

$$P(D_1) = 100\%, \qquad P(D_2) = 0\%$$

How can a photon entering two 50/50 beam splitters end up at the same detector every time?

What does "equal paths" mean? To understand how the two paths can interfere, suppose we have a specific configuration of the Mach-Zehnder interferometer in which the two arms have equal optical path lengths.

$$U : BS_1 \to M_1 \to BS_2$$

$$L : BS_1 \to M_2 \to BS_2$$

Deepening

Each arm connects the first beam splitter BS_1 to the second BS_2, passing through a mirror. The **geometrical path length** is simply the **total distance along that route**. For example:

$$L_U = \underbrace{d(BS_1, M_1)}_{\text{first segment}} + \underbrace{d(M_1, BS_2)}_{\text{second segment}}$$

$$L_L = d(BS_1, M_2) + d(M_2, BS_2)$$

Where L_U and L_L are the total geometrical path lengths of the upper and lower arms, respectively. **Equal geometrical paths** means:

$$L_U = L_L$$

Instead of geometrical path length, we can also consider the **optical path length**, which is the geometrical path length multiplied by the refractive index of the medium:

$$\text{Optical path length} = \text{OPL} = n \cdot L$$

So if both arms are in air under identical conditions:

- Having equal **geometrical lengths** means their **optical path** lengths are also **equal**.
- But if the arms contain **different materials**, their geometrical lengths could be equal while their optical path lengths **differ**.

Why do we care? Because the relative optical path length **determines the relative phase** accumulated along the two arms. For light of vacuum wavelength λ_0, the propagation-induced phase difference is:

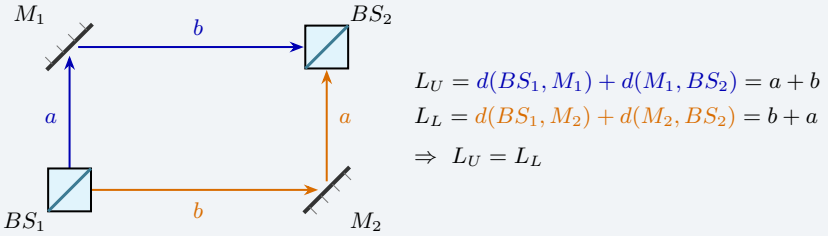
$$\Delta\phi = \frac{2\pi}{\lambda_0}\left(\text{OPL}_U - \text{OPL}_L\right)$$

Therefore, if the optical path lengths are equal:

$$\Delta\phi_{\text{propagation}} = 0$$

The two contributions accumulate the same phase during propagation. The beam splitters can still introduce additional relative phases, which determine how interference appears at the detectors.

(a) Geometrical path length



(b) Optical path length OPL $= nL$ (arms unrolled)

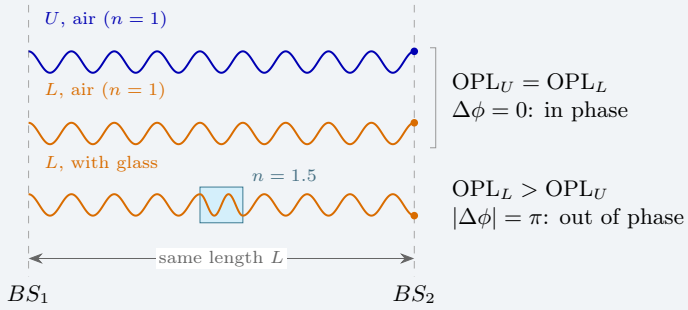


Figure 15: What "*equal optical paths*" means. (a) The geometrical length of an arm is the sum of its two segments; in the rectangular interferometer both arms cover one side a and one side b, so $L_U = L_L$. (b) The phase a wave accumulates depends on the optical path length $\text{OPL} = nL$. With both arms in air the two contributions reach BS_2 with the same phase ($\Delta\phi = 0$). A glass slab of index $n = 1.5$ in one arm shortens the wavelength inside it to λ_0/n, so that arm fits extra oscillations into the same geometrical length: here half a cycle, giving $|\Delta\phi| = \frac{2\pi}{\lambda_0}\,|\text{OPL}_U - \text{OPL}_L| = \pi$.

What happens at the second beam splitter? **Each final detector receives two possible quantum contributions**, one associated with the upper arm and one with the lower arm. At BS_2, these contributions **interfere**.

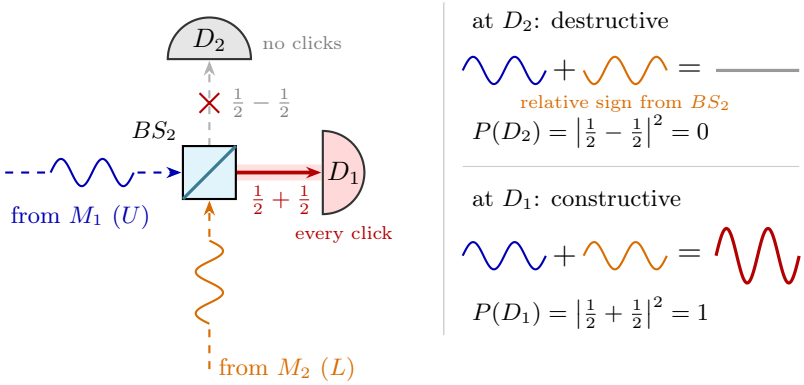


Figure 16: What happens at the second beam splitter for equal optical paths. Each output port of BS_2 receives two contributions, one from the upper arm U (blue) and one from the lower arm L (orange). Both reach BS_2 in phase; towards D_1 they **reinforce** $(\frac{1}{2} + \frac{1}{2})$, while towards D_2 the beam splitter gives them a relative minus sign and they **cancel** $(\frac{1}{2} - \frac{1}{2})$. The waves are a conceptual picture of adding or cancelling amplitudes, not electromagnetic fields measured at the detectors. Which detector gets all the photons depends on the phase conventions of the interferometer.

Look the figure above.

- At D_1, the two amplitudes **reinforce** each other.
- At D_2, the two amplitudes **cancel** each other.

Using the illustrative amplitudes on page 46:

$$P(D_1) = \left|\frac{1}{2} + \frac{1}{2}\right|^2 = 1 = 100\%$$

$$P(D_2) = \left|\frac{1}{2} - \frac{1}{2}\right|^2 = 0 = 0\%$$

Therefore, all photons are directed toward D_1 and none toward D_2. Obviously, which detector receives all the photons depends on the interferometer's phase conventions and configuration. Equal optical paths alone do not universally select the detector named D_1.

Why is this result so surprising? Consider our **classical prediction**:

$$P(D_1) = P(D_2) = \frac{1}{2}$$

Instead, the quantum prediction is:

$$P(D_1) = 1, \qquad P(D_2) = 0$$

This is simply the surprising part, because the two beam splitters are individually 50/50 devices (i.e., each has a 50% chance of transmitting or reflecting a photon), yet their combination can produce a 100/0 distribution (i.e., all photons going to one detector and none to the other). ***But why?*** Because the contributions from the two paths interfere instead of behaving like independent random choices.

If we know the detector, why don't we know the path? Suppose we have adjusted the interferometer to obtain:

$$P(D_1) = 1, \qquad P(D_2) = 0$$

We might reason: "*if we can predict exactly where the photon arrives, shouldn't we also know which path it followed?*". No, because these are two different measurements. Consider the following distinction:

- ***Where will the photon be detected?*** In this configuration, we can predict D_1 with certainty.
- ***Did the photon follow the upper arm or lower arm?*** Our experiment does not measure that. Both path contributions are involved in producing the interference at the final detector.

We therefore cannot conclude, for example, that the photon followed the upper arm just because it was detected at D_1. Also, we shouldn't simply imagine that it secretly made an ordinary independent classical path choice. That is the model whose predictions failed. The experiment establishes a definite final detection outcome, not a definite classical trajectory.

References

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